

HELIOLITE · REQUIREMENTS

HelioLite requirements

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DOCUMENT CONTROL

Document ID	HLT-REQ-001	Version	0.8
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heliolite		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Values from HLT-CAL-001 v0.5 (aluminum latch bracket, EPDM edge channel, two-lift fitting, BOM lines 1, 9 and 17); no target or status changed
0.7	2026-10-02	Amish Chadha	R13 note on the aluminum latch bracket and edge guard decided on 2026-10-02; no target or status changed
0.6	2026-10-01	Amish Chadha	Status from HLT-CAL-001 v0.4 for the constructable design (HLT-DDR-003); R15 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	R15 budget raised to \$430 (HLT-DDR-001 D4); R5 allows the decided anemometer (D2); status column replaced by the TRL 3 results of HLT-CAL-001
0.2	2026-09-25	Amish Chadha	

First measurable requirements for TRL 2, with status against the concept estimates

0.1	2026-09-24	Amish Chadha	Initial scaffold
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HelioLite requirements

These are the requirements for the concept, checked by calculation at TRL 3 in HLT-CAL-001 v0.5, for the constructable design of HLT-DDR-003 with the decisions of 2026-10-02 carried in. Targets are proposals, not yet validated with users. "Met" means met on paper only. On 2026-09-25 Amish decided to raise the budget to \$430 (HLT-DDR-001, D4), so R15 reads \$430, and R5 names the weather anemometer decided in D2 as allowed. Later the same day he accepted the TRL 3 recommendations (HLT-DDR-002): R13 is relaxed from 10 kg to 13 kg (N3), R7 now counts hands-on time spread over one clear day (N4), and R10 now allows the beam to move only downward during a stow (N5). On 2026-09-26 Amish approved a budget top-up to \$455 (HLT-DDR-002 v0.2, P1), so R15 read \$455. On 2026-10-01 Amish set budgets as hypothetical value-engineering targets ("the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"), so R15 now names the \$455 value-engineering target and its status is reported against it. No other target has changed.

The **design point** used throughout is a clear winter day at 45° N, DNI 800 W/m², a mirror-to-target distance of 10 m and an angle of incidence on the mirror of about 32° (cosine factor 0.85).

Table 1. Requirements and TRL 3 status (tags refer to lines printed by docs/04-calcs/sizing.py).

ID	Requirement	Target	Status (HLT-CAL-001)	Verification
R1	Redirected sunlight at the target	200 W or more arriving at the outside of the target glazing at the design point	Met: 207 W [A2]	Optical calculation (done)
R2	Useful daylight in the room	Raise the mean illuminance of a 12 m ² room by 300 lx or more at the design point	Met: 492 lx, utilization 0.4 assumed [A3]	Illuminance calculation (done)
R3	Daily energy delivered	0.5 kWh or more through the glazing on a clear winter day	At risk: 0.49 kWh on 21 Dec and 0.98 kWh on 1 Feb at reference site A; 0.40 kWh at site B on 21 Dec [B2, B3]	Hourly sun-path calculation (done); siting survey
R4	Pointing accuracy	Beam center within 0.5° of the target center (about 90 mm at 10 m), with winds up to 8 m/s	At risk: 0.30° typical and 0.50° at the 95th percentile, with four-point calibration and the decided preload springs; 1.04° without preload [D5, D6]	Error budget and calibration simulation (done); later spot log
R5	Sensorless tracking	Sun position from time and location only; no sun sensor or camera; axis homing switches and a weather anemometer allowed	Met by design	Design review
R6	Timekeeping	Clock error 60 s or less after 6 months without a network time source	Met: 31.5 s (DS3231, ±2 ppm) [D1]	Datasheet check
R7	Calibration	One person sets up and calibrates from a phone in 30 min or less of hands-on time, spread over one clear day, with no survey instrument (reworded 2026-09-25, HLT-DDR-002 N4)	Met: 19 min hands-on for four points over about 4 h [G1, G2]	Task analysis (done)
R8	Working range	Target distance 3 to 20 m; mirror normal azimuth ±135° about the target direction; elevation of the normal -90° (face-down stow) to +90°	Met mechanically; spot plus 0.5° error fits a 1.0 m window up to 16.8 m [A6, K1]	Geometry check (done)
R9	Wind	Hold R4 up to 8 m/s; survive 35 m/s (126 km/h) gusts in stow without damage	Met: the stowed moment (24.3 N·m, assumed coefficient) is carried by the stow stop and latch, designed for 48.6 N·m, not by the gearbox [E7, E8]	Wind load calculation (done); gearbox and pad data
R10	Safe beam when stowed, faulted or unpowered	Beam directed only at the target or the ground within 3 m of the mast when stowed; during a stow the beam may move only downward, toward the ground (clarified 2026-09-25, HLT-	Met: stow in 23 s on the supercapacitor reserve with 2.1x energy margin, beam moving only downward; latch engages without power [F1, F2, F4]	Calculation (done); later fault test

ID	Requirement	Target	Status (HLT-CAL-001)	Verification
		DDR-002 N5); stow face-down within 60 s of a fault or power loss		
R11	Electrical safety	Only 12 V DC (SELV) outdoors; electronics enclosure IP65; mains adapter indoors and listed	Met by design	Design review
R12	Outdoor life	Operate from -20 to +45 °C; UV-stable plastics (ASA or better); corrosion-resistant mast; 10-year mirror life with cleaning	Not verifiable at TRL 3	Materials review; later exposure test
R13	Mass and install	13 kg or less on the mast top (relaxed from 10 kg, decided by Amish 2026-09-25, HLT-DDR-002 N3); installed by two people with hand tools in 4 h or less	Met, very thinly: 12.99 kg on the mast top, 0.01 kg margin on assumed masses, with the aluminum latch bracket (0.08 kg less) and the EPDM edge channel (0.16 kg) decided on 2026-10-02; fitted in two lifts of 4.48 kg and 6.34 kg [H3]; install time not verifiable at TRL 3. If the weighed head exceeds 13 kg, R13 is relaxed slightly rather than the guard dropped (decided 2026-10-02)	Mass estimate (done)
R14	Standby power	3 W or less average from the 12 V supply	Met: 0.60 W [I1]	Power budget (done)
R15	Cost	Parts at or under the \$455 value-engineering target (a hypothetical control target; raised from \$400 to \$430 by Amish on 2026-09-25, then to \$455 by a top-up he approved on 2026-09-26)	Over the value-engineering target by \$44: estimated cost of the constructable design \$499 [J1]	Priced BOM (bom/ bom.csv)

■ Requirements not met, at risk or over the target

- **R15 (cost) over the value-engineering target by \$44.** The parts that make the design constructable (HLT-DDR-003) take the estimate from \$451 to \$492, and the edge channel and aluminum latch bracket decided on 2026-10-02 to \$499; the target is unchanged, and the design decisions register lists the cost drivers and savings worth trying.
- **R3 (daily energy) at risk.** Site dependent: site A now falls 2 % short on the solstice (0.49 kWh, after the edge channel's 3 mm front lip trimmed the clear aperture) and meets the target by February; missed at site B.
- **R4 (pointing) at risk.** 0.30° typical, but the 95th percentile of four-point calibration sits at the 0.5° limit; it relies on the decided preload springs against the gearbox's 1° backlash.
- **R12 (outdoor life) not verifiable at TRL 3.**
- **R13 (mass)** is met with only 0.01 kg of margin on assumed gearbox, spring and solenoid masses.

■ Assumptions

- Mirror area 0.36 m^2 ($600 \times 600 \text{ mm}$), clear aperture 0.353 m^2 inside the edge channel's 3 mm front lip, solar-weighted reflectance about 0.93 when clean and 0.95 soiling factor (combined 0.88).
- Double glazing transmits about 75 % of the beam at normal incidence, less at oblique incidence (HLT-CAL-001).
- Luminous efficacy of direct sunlight about 95 lm/W; a utilization factor of 0.4 for sunlight bounced from a light ceiling or a diffuser into a 3 x 4 m room.
- R3 is checked by an hourly clear-sky model at three illustrative sites at 45° N (HLT-CAL-001, section B). The TRL 2 shortcut of six useful hours at a mean DNI of 650 W/m^2 and a mean cosine factor of 0.80 gives 0.73 kWh and is no longer used.